W2 todo · Oct 12–18

☐	**Catherine & Luce**	Bring table to germination room
☐	**Catherine & Luce**	Prepare Flood Rack item list
☐	**Catherine & Luce**	Prepare Greenhouse Table item list

Stage 2 · rest zone floor, power + collector

☐	**Catherine & Luce (Michel)**	Rack power: all 2 West runs pulled
☐	**Catherine & Luce (Michel)**	Rack power: all 2 West runs pulled
☐	**Catherine & Luce (Michel)**	Seal the wire holes -- electrical paste + silicon
☐	**Guillaume (Michel)**	Wire outlets · west + north
☐	**Michel**	Add the 3" tee for the germination room · before the trenches are filled
☐	**Catherine & Richard**	Fill trench with dirt/sand
☐	**Michel & Richard**	West + North-West collector
☐	**Richard**	Dig Greenhouse waste line
☐	**Richard**	Dig Island trench

Stage 3 · 8 existing racks flooding → first sowing

☐	**Luce**	Rack wiring: replicate the sample
☐	**Guillaume**	Draw Fill lines
☐	**Luce & Catherine**	Fill lines
☐	**Catherine**	Switch timers + set timers week1 so that there's no manipulation needed
☐	**Guillaume**	Sample: flood rack + downpipe
☐	**Luce & Catherine**	Tray ports, 40 existing-rack trays
☐	**Luce & Catherine**	Drain-drop pre-assembly, 40

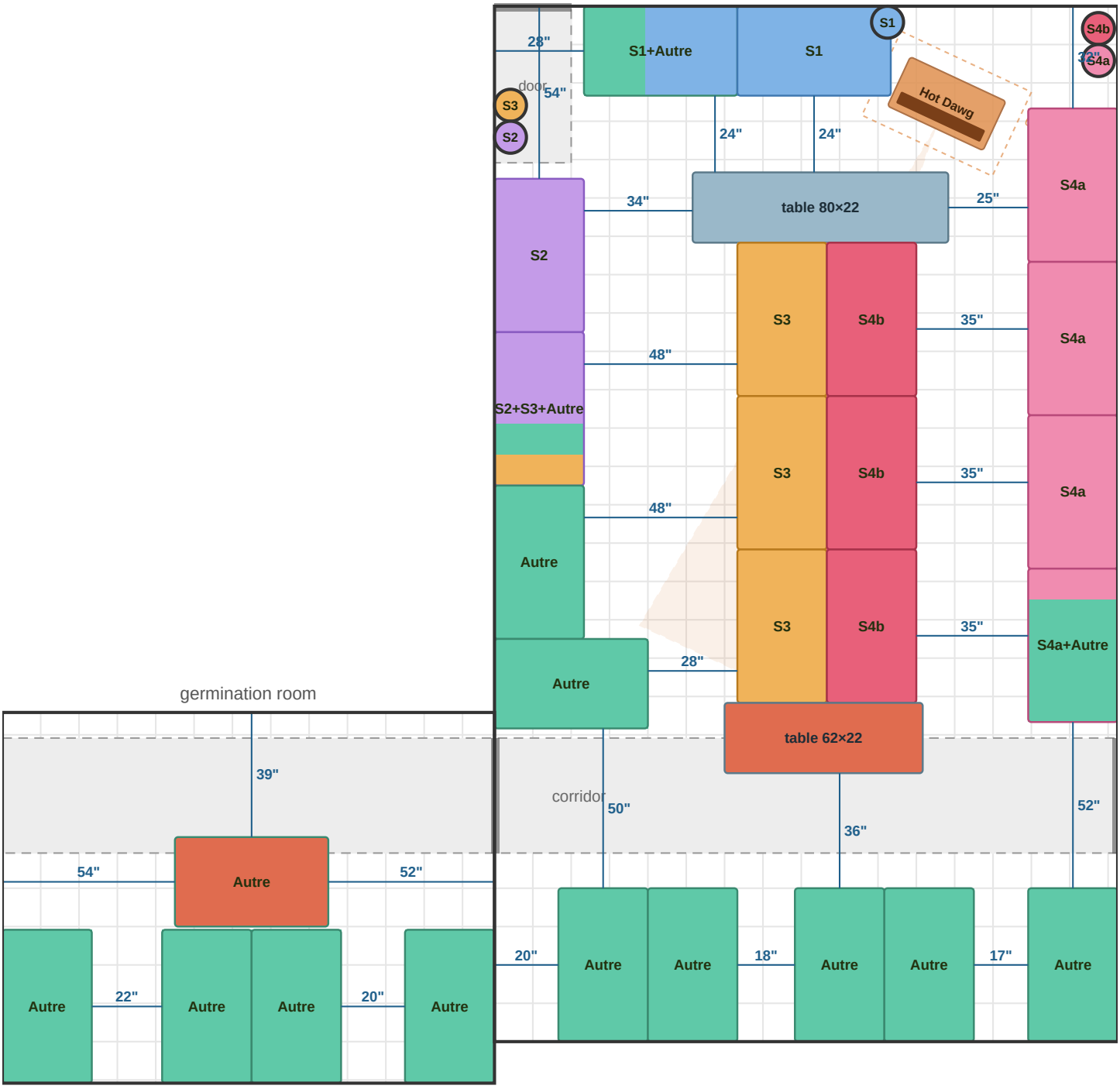
Stage 4 · greenhouse table area clear

☐	**Guillaume**	Sample: greenhouse table + 3 main lines ---- after the flood rack sample

Nursery room plan

S1 S2 S3 S4a S4b Autre (other) blocked

stripes = shelves, top → bottom · ○ fill valve per zone · 26 racks, 520 trays



Sample · 1 rack

Flood rack

Flood rack sample — parts to find (1 rack, 5 trays)

Indoor Farmer

☐		Grow Rack Shallow Rack Tray 45,5 × 25,5 × 4,5 po	**5**
☐		Active Aqua fill/drain fitting ¾	**10**
☐		Active Aqua screen fitting	**10**
☐		Active Aqua extension fitting	**10**
☐		Active Aqua vinyl ¾ ID, 100 pi roll	**~5 m**

U.S. Solid

☐		U.S. Solid motorized ball valve ¾ in SS, 3 indicator lights, 9–36 V, 2-wire auto-return, NO	**5**

Dubois

☐		Adaptateur mâle ¾ MPT × ¾ insert	**20**
☐		Valve barbée 0,18 po pour micro-tube	**5**
☐		Coude ¼ barbé	**5**

RONA

☐		IPEX tuyau ABS 1½ po × 12 pi	**1**
☐		IPEX té sanitaire ABS 1½ po F×F×F	**10**
☐		IPEX réduction ABS 1½ × ¾ po mâle × FPT	**10**
☐		IPEX coude 90 extra-long ABS 1½ po F×F	**3**
☐		Aqua-Dynamic manchon flexible PVC 1½ × 1½ po, colliers inox	**2**
☐		IPEX té sanitaire ABS 3 × 3 × 1½ po	**1**
☐		Oatey ciment ABS jaune 473 mL	**1**

Home Depot

☐		Commercial Electric attaches zip UV 8 in, 100-pack	**~10**

Amazon

☐		Collier inox 304 19–29 mm, 20-pack	**30 clamps**
☐		HydroMaxx flexible PVC 1½ in Sch 40, white, 25 ft	**~2 ft**

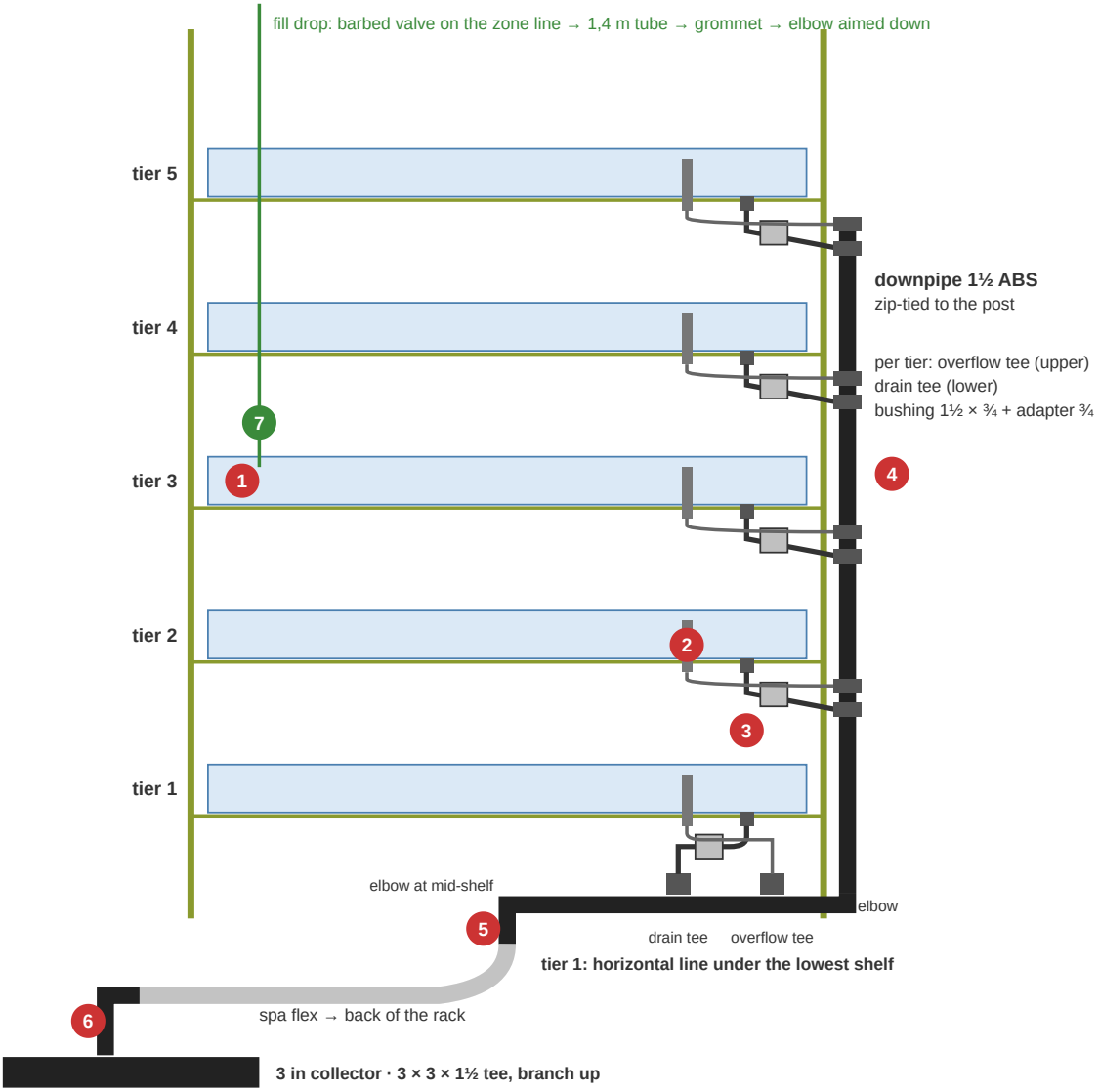
☐		Néoprène 1/16 × 2 in × 10 ft	**~10 pads**
☐		Vefhby neoprene strip 1/8 × 3 in × 10 ft, solid	**as needed**
☐		MIXC drip tubing ¼ in OD, 200 ft	**7 m**
☐		HORTIPOTS grommet ¼ po, 25-pack	**5**
☐		WAGO 221, 90-pack	**~10**

Not on hand

☐		Wire 18/2 (Orisha, not ordered)	**~8 m**

Sample — flood rack + downpipe

Spec: nursery/rack-water.md · tray-ports.md · tray-fill-drop.md · numbers = steps on the next pages



Flood rack + downpipe — parts, in order

1 · Tray + shims



Flood tray 45,5 × 25,5

→



Neoprene 1/16 in pads

→



Neoprene 1/8 in pads

Lift the flats → set the tray → pour 5 L → shim the high corners until all runs to the drain port → flats back.

2 · Ports



Fill/drain fitting ¾

→



Screen fitting (inside)



Fill/drain fitting ¾

→



Extension

→



Extension

→



Screen on top

Row 1 = drain port, flush with the floor. Row 2 = overflow standpipe ≈ 2,5 in.

3 · Drain drop + overflow drop



Drain port →



Vinyl ¾ ID →



Inox clamp →



Adapter ¾ MPT × insert →



Motorized valve ¾ →



Adapter ¾ MPT × insert →



Vinyl ¾ ID →



Adapter in the drain tee



Overflow port →



Vinyl ¾ ID →



Inox clamp →



Adapter in the overflow tee

Téflon on every thread. One inox clamp per barb.

4 · Downpipe, per tray tier



ABS 1½ pipe →



Overflow tee 1½ (upper) →



Bushing 1½ × ¾ SPG × FPT →



Adapter ¾ MPT × insert →



Drain tee 1½ (lower) →



Bushing 1½ × ¾ SPG × FPT →



Adapter ¾ MPT × insert →



Zip ties → rear post →



ABS cement

Glue the tees per tray tier, branches toward the tray; bushings glued in the branches; zip-tie the downpipe to the rear post.

5 · Tier 1 · horizontal line under the lowest shelf



Downpipe foot → Long 90 1½ → Stub → Overflow tee 1½ → Stub



Drain tee 1½ (branch up) → Stub to mid-shelf → Long 90 1½

Tier-1 drain + overflow drops land on these 2 tees.

6 · To the collector



Long 90 at mid-shelf → Stub → Flexible coupling → Spa flex 1½ → back of the rack → Flexible coupling



Stub → Long 90 1½ → Stub → Collector tee 3 × 3 × 1½, branch up → 3 in collector

7 · Fill drop + valve wiring



Zone line ¾ PE

→



Barbed valve (punched in)

→



Tube ¼ in, 1,4 m

→



Grommet (3/8 in hole)

→



Barbed elbow, aimed down

Wire 18/2 from the contactor

→



WAGO 221 splice

→



Motorized valve, in parallel

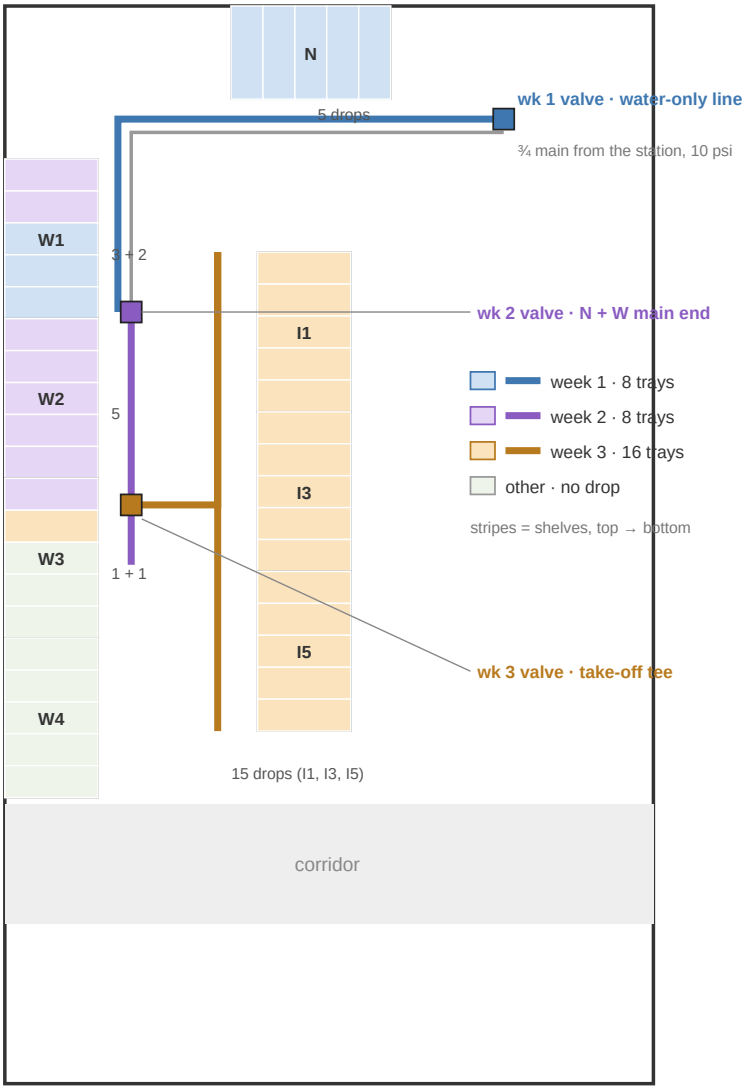
Hot-water dip to seat the tube on the barbs. Zip-tie the tube to the shelf.

Weeks 1 · 2 · 3

Fill lines

Plan — week 1 · 2 · 3 nursery fill lines

Spec: nursery/water-supply.md § Zone lines · shelf weeks from the default room plan · line routes schematic



Fill lines — parts, in order

1 · Week 1 · water-only line


Tee in the bib PE

→


Adapter ¾ MPT × insert

→


PSR-2 10 psi

→


Adapter ¾ MPT × insert

→


PE ¾, 15 ft

→


Reducing adapter 1 MPT × ¾ insert

→

Orisha fill valve 1 in (week 1)

Own 10 psi regulator: plain water, no fertigation.

2 · Each zone line (weeks 1, 2, 3)


Reducing adapter 1 MPT × ¾ insert

→

Orisha fill valve 1 in

→


Reducing adapter 1 MPT × ¾ insert

→


PE ¾ along the racks

→


Adapter ¾ MPT × insert (line end)

→


Cap ¾ FPT, unscrew to flush

Week 2 valve on the north + west main end · week 3 valve on a take-off tee.

3 · Turns + take-offs


Elbow 90 insert ¾

→


Tee insert ¾

→


Inox clamp, 1 per barb

Téflon on the threads.

4 · One drop per tray



Zone line ¾ PE

→



Barbed valve, punched 3 mm

→



Tube ¼ in, 1,4 m

→



Grommet, 3/8 in hole

→



Barbed elbow, aimed down

Week 1: 8 drops · week 2: 8 · week 3: 16.

Sample · 1 table

Greenhouse table

Greenhouse table sample — parts to find (1 table)

Indoor Farmer

☐		Active Aqua fill/drain fitting ¾	**2**
☐		Active Aqua screen fitting	**2**
☐		Active Aqua extension fitting	**2**
☐		Active Aqua vinyl ¾ ID, 100 pi roll	**~1 m**

Trimleaf

☐		xTrays Classic flood table 3×6, white, Canadian ABS	**1**

Dubois

☐		Adaptateur mâle ¾ MPT × ¾ insert	**2**
☐		Valve barbée 0,18 po pour micro-tube	**1**
☐		Coude ¼ barbé	**1**
☐		PE 1½ × 100 pi	**header section**

☐		Bouchon insertion 1½ po	**1**

RONA

☐		IPEX té sanitaire ABS 2 × 2 × 1½ po F×F×F	**1**
☐		IPEX té sanitaire ABS 1½ po F×F×F	**1**
☐		IPEX tuyau ABS 2 po × 12 pi	**1**
☐		IPEX tuyau ABS 1½ po × 12 pi	**1**
☐		Oatey ciment ABS jaune 473 mL	**1**

Home Depot

☐		LESSO ABS flush bushing 1½ × ¾ in SPG × FPT	**2**
☐		Commercial Electric attaches zip UV 8 in, 100-pack	**~6**

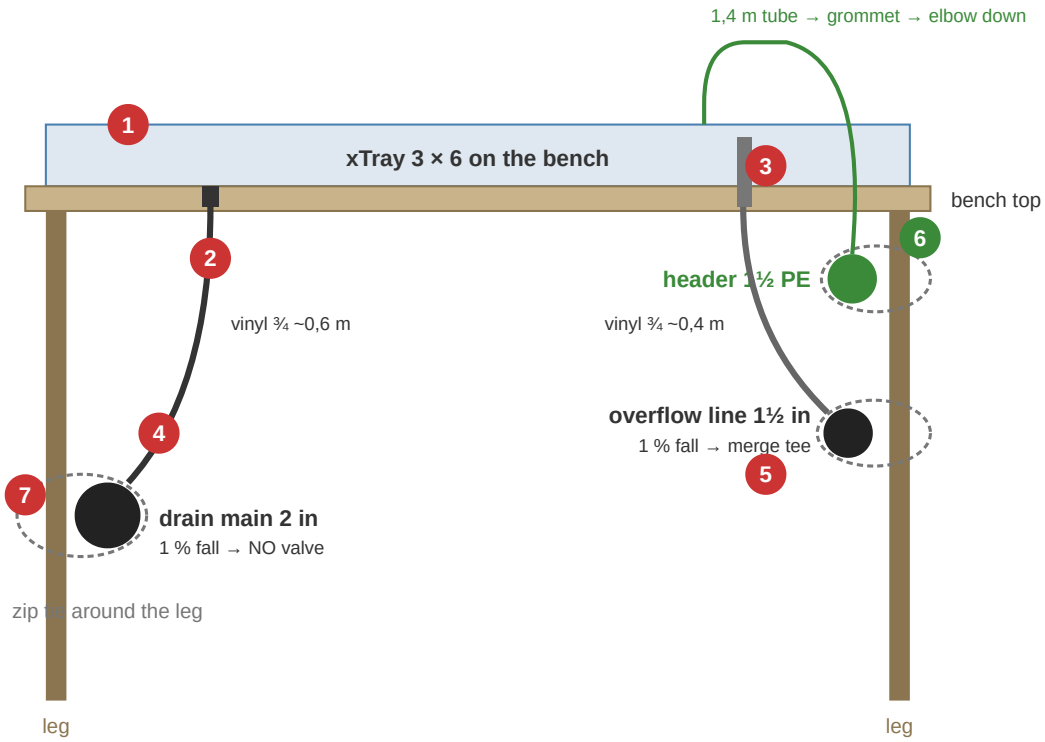
Amazon

☐		Collier inox 304 19–29 mm, 20-pack	**4 clamps**
☐		MIXC drip tubing ¼ in OD, 200 ft	**1,4 m**

☐		HORTIPOTS grommet ¼ po, 25-pack	**1**

Sample — greenhouse table: tray, drops + 3 main lines

Spec: greenhouse/table-water.md · waste.md · water-supply.md



tee on each main lined up under its port · branch up, sweep toward the valve / merge tee

Greenhouse table — parts, in order

1 · Table + ports



xTray 3 × 6 on the bench

→



Drain fitting ¾

→



Screen (inside)



Overflow fitting ¾

→



Extension

→



Extension

→



Screen on top



Fill grommet, 3/8 in hole

Drain port at the moulded spot nearest the drain main.

2 · Drain drop



Drain port

→



Vinyl ¾, ~0,6 m

→



Inox clamp

→



Adapter ¾ MPT × insert

→



Bushing 1½ × ¾ SPG × FPT

→



Tee 2 × 2 × 1½, branch up

→



Drain main 2 in

Téflon on the threads.

3 · Overflow drop



Overflow port



Vinyl ¾, ~0,4 m



Inox clamp



Adapter ¾ MPT × insert



Bushing 1½ × ¾ SPG × FPT



Tee 1½, branch up



Overflow line 1½ in

4 · Fill drop



Header 1½ PE



Barbed valve, punched 3 mm



Tube ¼ in, 1,4 m



Grommet



Barbed elbow, aimed down

5 · Mains on the legs



Drain main 2 in, 1 % fall



Overflow 1½ in, 1 % fall



Header 1½ PE



Zip ties around the table legs

Mark the 1 % fall on the legs first.